$\lim_{x \to a} f(x) = \pm\infty$

Vertical Asymptote

if you get a $\frac{\text{nonzero}}{\text{zero}}$,

answer is $\pm\infty$

$$\lim_{x \to 2} \frac{x^2 + 4x - 3}{x - 2} = \frac{9}{0} = \pm\infty$$

$$\lim_{x \to 1^+} \frac{x^3 + 4x}{1 - x} = \frac{5}{0} = -\infty$$

$-$

$\lim_{x \to \pm\infty} f(x) =$

Horizontal Asymptote

1) $n < d$, bottom is rising faster, 0
2) $n = d$, same rate, $\frac{c}{c}$
3) $n > d$, num fast, $\boxed{\pm\infty}$

radicals & absolute value,

[illegible]

$$\lim_{x \to \infty} \frac{\sqrt{5x^6 + 3x}}{2x^3 - 4} = \frac{\sqrt{5}}{2}$$